

Mathematics Questions - Algebra

Subject: Mathematics | Chapter: Algebra | Difficulty: medium

Question 1: Solve the equation: $3x^2 - 7x + 2 = 0$

Options:

- A) $x = 2, x = 1/3$
- B) $x = -2, x = -1/3$
- C) $x = 2, x = -1/3$
- D) $x = -2, x = 1/3$

Difficulty: 4/5

Question 2: Simplify the expression: $(2x + 3y)^2$

Options:

- A) $4x^2 + 9y^2$
- B) $4x^2 + 12xy + 9y^2$
- C) $4x^2 + 6xy + 9y^2$
- D) $2x^2 + 6xy + 3y^2$

Difficulty: 3/5

Question 3: Solve the system of equations: $x + y = 5$ and $x - y = 1$

Options:

- A) $x = 3, y = 2$
- B) $x = 2, y = 3$
- C) $x = 4, y = 1$
- D) $x = 1, y = 4$

Difficulty: 3/5

Question 4: What is the slope of the line $2x - 4y = 8$?

Options:

- A) $1/2$
- B) 2
- C) $-1/2$
- D) -2

Difficulty: 3/5

Question 5: If $f(x) = x^2 + 2x - 3$, find $f(3)$

Options:

- A) 6
- B) 12
- C) 0
- D) 18

Difficulty: 2/5

Question 6: Solve for x: " $(x + 5) = 4$

Options:

- A) 11
- B) 16
- C) 21
- D) 19

Difficulty: 3/5

Question 7: Factor the expression completely: $4x^3 - 16x$

Options:

- A) $4x(x^2 - 4)$
- B) $4x(x - 2)(x + 2)$
- C) $4x(x - 2)^2$
- D) $4x^2(x - 4)$

Difficulty: 4/5

Question 8: Find the x-intercept of the line $y = 3x - 6$

Options:

- A) (2, 0)
- B) (0, 2)
- C) (-2, 0)
- D) (0, -2)

Difficulty: 2/5

Question 9: Simplify the rational expression: $(x^2 - 9) / (x - 3)$

Options:

- A) $x + 3$
- B) $x - 3$
- C) $x - 9$
- D) $x + 9$

Difficulty: 4/5

Question 10: Solve the inequality: $2x + 5 > 11$

Options:

A) $x > 3$

B) $x < 3$

C) $x > 8$

D) $x < 8$

Difficulty: 2/5

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